

[Home](#) › [My Courses](#) › [Portfolio Management](#) ›[M4: Behavioral Finance and Applications to Portfolio Theory](#) › Lesson Notes

Lesson Notes

MODULE 4 | LESSON 1

BEHAVIORAL FINANCE

Reading Time	60 minutes
Prior Knowledge	Traditional portfolio theory and utility theory.
Keywords	behavioral finance, bounded rationality, satisfice, prospect theory, cognitive biases, emotional biases.

So far, we have viewed investors and economic agents in general as rational economic beings. Rational economic beings are assumed to have perfect information and are capable of acting in a perfectly rational manner all of the time. But is this really the case? Over the years, psychologists have questioned this assumption, and new paradigms have begun to emerge. This has led to a shift from traditional finance to behavioral finance, which has implications for how people feel, think, and act when confronted with economic decisions. This will be the subject of Module 4, where we consider the needs and implications of shifting from the world of traditional finance to that of behavioral finance.

1. From Traditional Finance to Behavioral Finance

What we have done so far in Modules 1, 2, and 3 can be referred to as **traditional finance (TF)**. What do we mean by that? It is very simple: in TF it is assumed that people are rational and risk-averse and that they act to maximize their own utility. In particular, the latter point means that individuals do not consider items like social values, unless these values can contribute to increase their own personal utility, that is. The typical individual in TF is considered a rational agent, called a **rational economic man (REM)**. The traits that characterize an REM are:

1. possessing perfect information, and
2. acting in a perfectly rational manner .

What does it mean to act in a perfectly rational manner? It means that individuals with a limited budget will select a mixture of goods and services to maximize their utility function. A utility function is a system of preferences that conform to the following 4 axioms:

- Given the choice among two portfolios or goods, A and B, an individual is always in a position to say whether they prefer A or B or is indifferent between the two (**completeness**);

- If portfolio A is preferred to B and B is preferred to C, then A is also preferred to C (**transitivity**);
- If portfolio A and B are mutually exclusive and A is preferred to B, then for any additional portfolio C and a number $\gamma \in (0, 1]$, $\gamma A + (1 - \gamma)C$ is preferred to $\gamma B + (1 - \gamma)C$.
- If A is preferred to B and B is preferred to C, there is a combination of A and C for which the individual is indifferent to B (**continuity**).

Note: The independence axiom *may* be dependent on the setting. An individual may like ice cream more than ketchup. But it is unlikely that they would like ice cream + fries more than ketchup + *fries*! And yet, this may not always be true.

The behavior of an REM leads to efficient markets, i.e., markets in which prices reflect all the relevant information available.

What happens when an REM gets a hold of new information? An REM has beliefs, and when presented with new information, they adapt their beliefs about the probabilities of uncertain events via Bayes' formula (in this sense, recall the Black-Litterman model as a sophisticated example). Hence, a REM must know the Bayes formula and be capable of performing the required calculations.

As it was already stated above, TF is also characterized by individuals who are risk averse. We know from Module 1, Lesson 3 that this means that their utility function is concave.

Behavioral Finance (BF), unlike TF, **does not assume** that:

- individuals are always risk-averse;
- individuals use Bayes' formula;
- individuals have perfect information;
- individuals act to maximize only their personal utility.

This does not mean that individuals cannot be REM, but they need not be all the time. Instead, the key assumptions of BF:

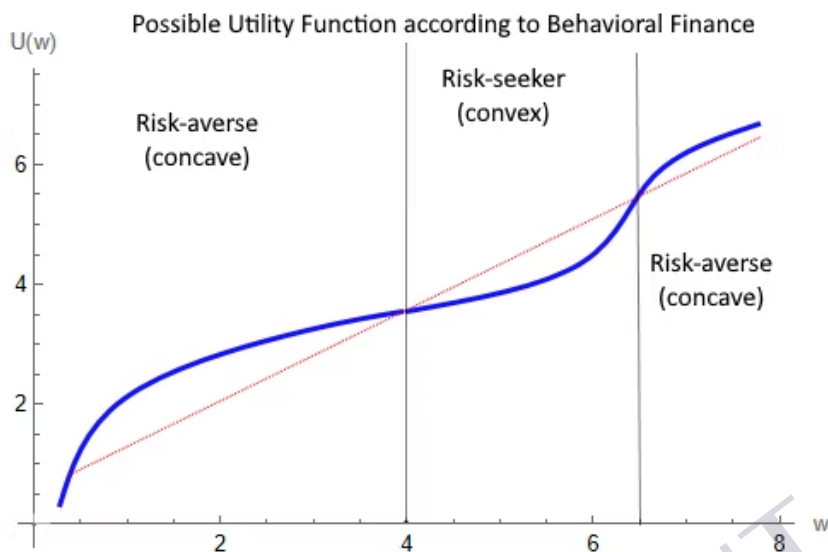
- individuals may not have access to full information;
- individuals may lack the mathematical ability to incorporate full or partial information in their decision making process;
- individuals may not be able to attain the highest utility because they are more interested in short-term goals as opposed to overall goals.
- individuals may consider philanthropy or just simple acts of kindness as useful while these have no place/value in the world of TF, which is ruled by self-interest.
- individuals' utility functions that are not always concave. Utility functions can be concave, convex (risk-seeker) or a mix of concave and convex.

In essence, the behavior of individuals in BF cannot be reduced to that of perfect calculators and their decision making is better explained by psychology.

2. From Utility Maximization to "Satisfice"

TF is based on utility theory and in particular concave utility functions (i.e., diminishing marginal utility). BF allows for more complex types of utility functions since it allows individuals to be both risk averse and risk seekers. In BF, an individual may exhibit a utility function like in Figure 1:

Figure 1: The Double Inflection Utility Function (Friedman and Savage)



In the first region, the individual is risk averse, i.e., they buy insurance in the language of utility theory we developed in Module 1. In the second region, the individual buys lottery tickets. In the third region, they are back to buy insurance. An REM would not behave like this. However, an individual driven by perceptions may decide that with wealth between 0.5 to 4 (in the picture of figure 1), they may need to protect it. When wealth is between 4 and 6.25, they may feel that there is enough wealth to take care of a certain amount of basic needs, and it is fine to gamble some to achieve a higher level of wealth still. Once wealth is above 6.25, the individual is back to being risk-averse because perhaps they have reached a very desirable level of wealth to fulfill their basic needs and some additional aspirational goals as well and feels that now wealth must be protected lest they lose the ability to fulfill such aspirational goals. It should be evident that we are assuming that individuals are driven by their own psychological perception of wealth in terms of goals that they can or cannot realize.

Is BF supported by evidence? The answer is yes. For instance when individuals suffer from anxiety, irritability, or depression they exhibit a certain type of behavior towards losses or risks. On the other hand when they feel euphoric they display a different attitude toward losses and risk. Some studies have pointed out that this may be a matter of our bodies producing serotonin or dopamine although the link is not accepted by everyone. It has also been pointed out that when an individuals anticipate rewards, they produce more dopamine and if they do not achieve certain results, then the dopamine level goes down.

The science that tries to establish a relationship between the behavior of investors and the way the brain works is called **neuronomics**.

TF has no explanation for asset bubbles, for instance. However BF can explain it. TF has no explanation for the equity risk premium puzzle: Rajnish Mehra and Edward Prescott in their 1985 paper found that the average real annual yield on equity in competitive pure exchange economies should be "a maximum of 0.4% higher than that on short-term debt." However, they found that

historically it stood at 6%. BF can provide some justifications for this, and there are more instances that highlight the fact that BF is a real thing and not just a nice theory.

Having said what we just did, it is useful to explain that the attitude towards BF is not uniform among practitioners and academics. To this purpose, Nassim N. Taleb wrote in *Statistical Consequences of Fat Tails*, on page 12, that:

"Examples of fields that are, practically, frauds in the sense that their results are not portable to reality and only serve to feed additional papers that, in turn, will produce more papers: Modern Financial Theory, econometrics (particularly for macro variables), GARCH processes, psychometrics, stochastic control models in finance, behavioral economics and finance, decision making under uncertainty, macroeconomics, and a bit more."

Clearly, Taleb kills many branches besides behavioral economics and finance in his statement. Nevertheless, several prescriptive approaches (i.e., approaches that are not derived by theories but by observing how investors actually make decisions) have been studied. We could call these approaches **heuristic approaches**, which were pioneered in the 1970s and 1980s by Kahneman and Tversky (see the References for their seminal 1979 paper in *Econometrica*). Daniel Kahneman, a psychologist, was awarded the Nobel Prize in Economic Sciences in 2002. Amos Tversky, a mathematical psychologist, died in 1996 but Kahneman acknowledged in an interview with the Stanford Report ([find the interview here](#)) that *"The prize ... is quite explicitly for joint work, but unfortunately there is no posthumous prize."* In 2017, Richard Thaler was also awarded the Nobel Prize in Economic Sciences "for work that had explored how human psychology shaped economic decisions."

In the beginning, many economists rejected behavioral finance as experiments lacking clear links to real life; however, later, the field has gained more acceptance. In a 2010 commentary ([link to commentary](#)), Kenneth Arrow, co-winner of the 1972 Nobel Prize for Economic Sciences wrote that:

"We see, by the way, that today psychology is invading economics—the whole field of behavioral economics. I believe that sociology should play more of a role in economics than it does. The way people behave in economics is partly influenced by how other people behave. It's easy to point out examples, but it's not so easy to construct a broad theory."

If REM is the typical individual for TF, the prototype individual of BF exhibits **bounded rationality**. Essentially, individuals have limits in their knowledge capacity and the assumption of perfect information must be removed. Individuals, instead of maximizing their utility, content themselves with satisfice, that is individuals operate with partial information and use heuristics instead of mathematical theories to process their information and transform them into decisions.

Example (satisfice). Say that you were hired for a job and need to relocate to a new city. Among other things, you will need to find an apartment or a house to live in, get utilities connected, find a bank, etc. Let's concentrate on finding a bank. As an REM, you would go all over town, find out everything you need to know about every single bank, and then decide which bank is the best one for you. Proximity may be taken into account in your utility function as much as interest rates, fees, and types of transactions the bank can execute for you. However, collecting all the information is quite a chore. The BF individual will probably visit a few banks and then decide based on that partial information that one of those rates offered is adequate and probably would not attach a lot of utility to obtaining more interest rates and going all over town for several more days. This is an example of

bounded rationality and satisfice. Same is true for the housing unit. Will you visit a certain number of units before deciding on one, or will you visit every single available unit before making your decision?

Example (satisfice). Max Schmelling has a new client, Jon, who is interested in becoming an antique car investor and is ready to spend \$200,000 from his portfolio to buy the first car for his collection. Shortly afterwards, the client visits an antique car auction not far from his home. Unfortunately, the auction had a limited number of cars meeting his requirements. He is subsequently drawn to one antique car in particular, even though it does not have several of the features he wanted. Nevertheless, after some consideration, he decides to purchase it anyway. and the car is now in storage for safekeeping. This is an example of *satisfice*.

Exhibiting bounded rationality should not be considered a pejorative form of rationality. Bounded rationality takes into account that it is often not practical to consider all possible outcomes or that computations may become too complex. In the contest of portfolio theory, while an REM will strive for the optimal solution, the "satisfice man" will be fine with a satisfactory solution, albeit suboptimal.

3. Prospect Theory

As already stated, bounded rationality removes the assumptions of perfect information and maximization of expected utility. Prospect theory goes one step further and replaces the assumption that individuals are risk averse with another assumption according to which individuals **exhibit loss aversion** instead. In particular, individuals fear losses more than they enjoy gains. In other words, **investors focus on change in wealth** rather than on wealth per se, and given a potential loss and gain of equal amounts, the increase in utility associated with the gain is smaller than the disutility associated with the potential loss. Note the subtle points: It is not just that economic agents have positive or negative change in values of utility from gain or losses; respectively, the change in the case of losses are larger than the changes associated with gains.

Experiments have shown that most individuals if offered a gamble with 50/50 odds of equal but opposite payoffs, they will not take the gamble. The only explanation for this is that they overreact to small probabilities and underreact to large probabilities, i.e., probabilities have a psychological component as well. A 1% probability of losing, for example, might be perceived as 10%, and a 99% probability of winning is felt as if it were, say, 90% instead.

Finally, individuals experience diminishing sensitivity applied to upward and downward perceptions and to evaluation of changes of wealth. This means that, given a set of possible outcomes $\{x_1, \dots, x_n\}$ with probabilities $p_1, \dots, p_n$, the utility function in prospect theory looks like:

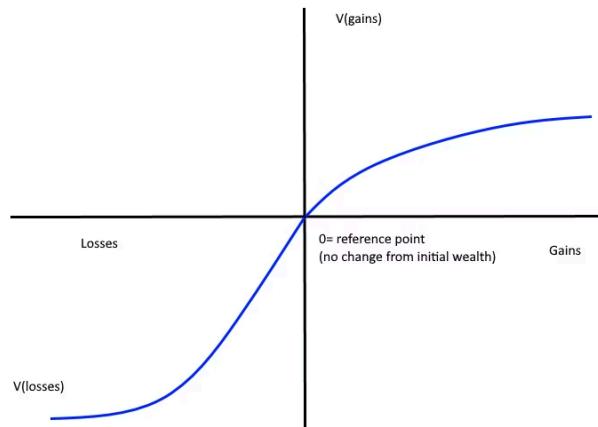
$$U(x_1, \dots, x_n) = w(p_1) \cdot p_1 \cdot V(x_1) + \dots + w(p_n) \cdot p_n \cdot V(x_n)$$

where $w(\cdot)$ is a weighting function assigned to probabilities. In TF, it would be $w(p) = p$, while in prospect theory, it can be something else.

Also, we assumed n possible outcomes, but the number of outcomes can be infinite. $V(\cdot)$ is a value function that is based on changes and not on wealth so that $V(x_i)$ represents the change in wealth if outcome x_i is the result.

To reflect the descriptions of prospect theory, the function V must look something like in figure 2, i.e., S-shaped and asymmetrical.

Figure 2: The Function $V(\cdot)$ is S-shaped and Asymmetric



Example. Consider the following situation in the context of prospect theory:

Outcome	value	probability	weight(probability)
x1	-100	3%	1.50
x2	-35	5%	1.20
x3	-1	25%	0.90
x4	30	30%	0.80
x5	90	37%	0.75

Then, the utility function can be calculated as

$$U = -100 \cdot 1.50 \cdot 3\% - 35 \cdot 1.2 \cdot 5\% - 1 \cdot 0.90 \cdot 25\% + 30 \cdot 0.80 \cdot 30\% + 90 \cdot 0.75 \cdot 37\% = 25.35.$$

This psychological approach warns us about the fact that investors fear losses and can become risk seekers in an attempt to avoid losses despite being risk averse when presented with gains.

Empirical studies have shown that:

- on average, if individuals are presented with a gamble where they have an equal chance to win \$100 or lose \$70, they will not take the gamble;
- on average, if offered a sure loss of \$75 or a gamble where they have 50/50 chance to win \$30 or lose \$200, they take the gamble even if its expected value is -\$85.

Prospect theory can explain why investors tend to concentrate on high-risk and low-risk investments and do not seem to have much of an appetite for medium-risk assets.

To understand why individuals behave the way they do, one has to dig into the psychological causes of these behaviors, which will be the subject of the next section.

In the meantime, Table 1 summarizes the key differences between TF and BF.

Table 1: Summary of the Differences between TF vs. Bounded Rationality and Prospect Theory

Traditional Finance	Bounded Rationality+Prospect Theory
Perfect knowledge	Limited knowledge
Utility is maximized	Satisfice
Decision making is fully rational	Limitations on cognitive ability when making decisions
Risk aversion	Loss aversion

4. Behavioral Biases of Individuals

Individuals are called upon to make thousands of decisions every day. Traditional finance's REM follows the fully rational steps. However, many individuals, when facing complex decision-making situations, often either lack the time, the ability, or both to follow an optimal course of action and are prone to take shortcuts and/or follow their intuition or choose to utilize simple rules of thumb. Furthermore, most individuals are also prone to let emotions and social influences affect their decision-making process.

An error is something that happens at random; a bias is something that is systematic and thus predictable. We cannot do much about unpredictable errors; however, BF tells us that individuals suffer from many biases. For convenience, they can be grouped in two main groups: **cognitive and emotional biases**.

- cognitive biases are the result of faulty reasoning and/or the inability of individuals to follow and/or understand complicated arguments (statistical techniques, processing information, memory errors, honest mistakes, etc.);
- emotional biases are caused by individuals' feelings and impulses or by following one's intuition and, as such, are not completely conscious.

Most individuals suffer from a combination of both groups. It is not always easy to distinguish what is what. However, we should make an attempt because a cognitive bias can be eliminated or reduced via better training or education; an emotional bias, on the other hand, is more difficult to remove and often has to be accommodated.

It would be a mistake to think of these biases as something completely theoretical. Real individuals do not behave like textbook individuals.

Example: Dealing with customers as a portfolio manager. Suppose you are assigned a customer, a 69-year-old woman who inherited a considerable amount of money from her late husband. Her late husband served as an executive for an oil company for over three decades and left her a considerable amount of wealth, mostly tied in oil stocks. This person comes to your office to discuss her portfolio. You explain to her that she is in a very good situation but that you are concerned that nearly 75% of her holdings are tied up in oil companies and that the oil industry is not expected to do particularly well over the next 2–3 years. Immediately, she shuts you down: "I would not be honoring the memory of my late husband if I were to sell the assets he left me, so these assets cannot be touched!" This is clearly a situation in which a person exhibits an emotional bias. It also appears that it would prove nearly impossible to remove it, since she feels so strongly about it. The best thing you can probably do is to try working with her instead; perhaps consider moving some assets around but within the oil industry sector. She accepts, as long as you keep her investments in oil companies. This is an example of moderating a bias. Had you insisted on

changing the composition of her portfolio dramatically or that she reduce her exposure to the oil sector, that action could have led to an unhappy customer or even to a customer firing the firm. Portfolio managers have to earn a living too.

These situations happen more often than you might think. In another instance, you may find yourself dealing with a 50-year-old man who created his own company and has done very well for himself. He has a sizeable portfolio consisting of mostly tech stocks. He has no problem changing the composition of his portfolio but has one demand: high returns and avoiding "boring investments like those his father makes." Again, this is another individual who exhibits emotional biases mostly. He was smart enough to understand your objections when you point out that his portfolio is very risky, but he feels that he knows when to get out of it if necessary. Again, you will be forced to attempt to moderate his biases by proposing tech stocks that you and or your analysts perceive as safer than several of the names he wanted to hold. Trying to correct his excesses would not sit well with a person who feels they can do anything.

Finally, a third case concerns a 60-year-old woman who is a senior executive in the fashion industry. She also has an excellent portfolio accumulated over a period of many years but is returning only 2–3% a year. You explain to her that, without increasing her risk much, she could achieve better returns. She becomes sort of frantic and explains to you that she is very worried about losing money as she plans to live off it in retirement and provide for her sick sister. However, after you showed her several simulations and she realizes that the additional risk is not substantial, she gives you the mandate to change the composition of 1/3 of her portfolio. This is a case of a person who had cognitive errors (she could not understand the simulations and what they meant) but once you explained them to her, she became open to the possibility of changing her portfolio's composition.

What is the moral of these short stories? One cannot always expect to be dealing with an REM and has to learn to deal with real people and their strengths and weaknesses, or they can lose a client. It's very possible that you, as a quantitative analyst, will end up in this position, and losing clients is not a good thing. These concepts are not just theoretical; they happen in real life.

The following tables list the most typical cognitive and emotional biases and provide general descriptions.

Table 2: Cognitive Biases

Type of bias	Description of bias
conservatism	information used to make the original decision/forecast is considered more relevant than new information
confirmation	looking for data that support one's beliefs and discounting information that contradicts one's beliefs
anchoring	reluctance to modify opinions or decisions, even when the information makes those conclusions unlikely. The individual is locked on certain ideas.
illusion of control	individual feels that they have control over outcomes when in reality they have little or none at all
hindsight	selective memory; i.e. individual remembers past events where she was right but forgets instances where she was wrong

Type of bias	Description of bias
framing	information is treated differently depending on how it is presented or obtained.
representativeness	assuming that an original classification and methodology works in all cases. Sometimes it is described as deducing too much from small samples of new information.
mental accounting	considering multiple investments as separate entities and not as an overall portfolio.
availability	memorable events are given more weight in changing the probability of events than less memorable ones.

Table 3: Emotional Biases

Type of bias	Description of bias
loss aversion	assigning more (negative) value to losses than (positive) value to gains of same amount
overconfidence	illusion of having better information than others or a superior ability to "read the markets"
self-control	lack of discipline, favoring short-term goals over long-term ones
self-attribution	belief that one's successes are due to their own character and abilities, rather than luck or situational factors
status quo	preferring one's environment and/or situation to new ones, described also as reluctance to change(s).
endowment	valuing what one has more than something they do not own
regret aversion	doing nothing for fear that decisions could turn out to be wrong

Example. Tina has just received new information regarding her investment in SuperStar Company. She forecasted a return for SuperStar of 10%. The new information seems to point to a potential annual return of 4%, instead. However, she is unwilling to use the new information to revise her forecast. Which particular bias is Tina displaying? How would your answer change if Tina decided that, despite the new information, she still believes a 10% return is achievable?

This is a clear case of conservatism bias because Tina is clearly assigning more weights to the original information used to forecast a 10% return rather than the new information. In the second case, the fact that, despite the new information, she feels that her original forecast should still stand resembles more an instance of anchoring bias as she does not feel like she should modify her original forecast.

Example. Jayne set up her initial asset allocation in her defined contribution plan over 20 years ago. She was then 25 years old and fresh out of graduate school with limited financial education. She decided to put an equal amount into each asset. Over time, she has increased her contribution to the plan. Her portfolio did well and at age 47, she has a portfolio valued at \$1.3 million while her asset allocation has not changed much over the period. What bias does Jayne display? Could it be corrected or should it be just moderated?

There is nothing particularly outrageous about Jayne's portfolio. The only elements we are given is that she invested the same amount in each asset over 20 years ago and the weights in her portfolio have not changed since. This makes for a textbook case of status quo bias since the investor leaves the weights unchanged and does not change them in response to changing market conditions. Since status quo bias is an emotional bias, it is difficult to correct, and it is more of a candidate for moderation.

Example. Max inherited substantial wealth from his family a number of years ago. He is now 55 and devotes his time to philanthropic activities. He comes to Jonas as a new client referred by a mutual friend. During their first meeting, Jonas has noticed that the majority of Max's wealth is concentrated in the stock of a large industrial company that his grandfather cofounded. Jonas explains his concerns about risk and advises Max to replace this stock with a more diversified and efficient portfolio. Max is very agreeable. However, when Jonas warns Max that changing the portfolio composition requires selling a large amount of assets and that this will trigger a large capital gains tax, Max becomes reluctant to sell. Which type of bias is Max most likely displaying?

This is a case of framing bias. In fact, Max's attitude towards selling the position to invest in a new portfolio changes depending on how the decision is framed. Max is fine moving to a more diversified and efficient portfolio. However, he becomes reluctant to do so when presented with the tax consequences of selling the current stock position.

Example (Lessons from student's life). Mental and cognitive biases go beyond investing and are a reality of daily life.

- Many students fail their exams because they do not exercise sufficient self-control to study enough.
- Students exhibit overconfidence when, because they have passed previous exams, they feel they can also pass later exams without too much sweating.
- Students display representativeness bias when they assume that the way they studied for their Financial Markets class will work for their Derivatives class as well.
- Students exhibit self-attribution bias when they procrastinated studying before a hard assignment (self-control bias) and have now only a few days to work on it. Students then receive low scores and blame the fact that they merely ran out of time for the poor performance instead of recognizing their inability to organize their work or their lack of skills.

5. Conclusion

This lesson introduced behavioral economics and explained in what ways it is different from traditional finance. We also covered how the transition from one to the other implies leaves behind a strict notion of utility maximization in favor of the new concepts of satisfice and prospect theory.

After establishing the need to bring individuals' behaviors into the picture, we analyzed the most typical cognitive and emotional biases displayed by individuals that are responsible for the need to change the way we look at finance to some degree.

The next lesson will cover in more detail the elements of prospect theory, explaining how prospect theory is applied and some of the mathematical tools on which it is based.

References

- Kahneman, Daniel, and Amos Tversky. "Prospect Theory: An Analysis of Decision under Risk." *Econometrica*, vol. 47, no. 2, 1979, pp. 263–291.
- Mehra, Rajnish, and Edward C. Prescott. "The Equity Premium: A Puzzle." *Journal of Monetary Economics*, vol. 146, 1985, pp 145–161.
- Mehra, Rajnish, and Edward C. Prescott. "The Equity Premium in Retrospect." *NBER Working Paper*, no. w9525, 2003 (revised 2022).

